TP02: Working with Big Data Files (+5GB)

Big Data Analytics
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1 Objective

The objective of this practical work is to compare different methods for processing large datasets. The experiments focus on memory consumption, execution time, Pandas Chunking, and Dask DataFrame.

2 Dataset and Environment

The dataset used is NYC Taxi Fare Prediction from Kaggle. The training file contains 55,423,856 rows and 8 columns. The experiments were performed using Python and the Pandas, Dask, and psutil libraries.

3 Task 1: Traditional Pandas

In the first task, I used `pd.read_csv()` to load the entire CSV file into memory.

The operation completed successfully in 392.28 seconds. The reported process memory after loading was 9.36 GB. The mean of the `fare_amount` column was approximately 11.3450456.

No MemoryError occurred during this experiment. However, the method required more execution time than the other approaches.

4 Task 2: Pandas Chunking

In the second task, I processed the dataset in chunks of 100,000 rows. Each chunk was processed separately, and the sum and count of valid fare values were accumulated to calculate the global mean.

The execution time was 138.42 seconds, and the observed RAM usage was 0.12 GB. The calculated mean was approximately 11.3450456.

This method reduced memory usage because the complete dataset did not need to be loaded into memory at once.

5 Task 3: Dask DataFrame

In the third task, I used Dask DataFrame to process the dataset in partitions. The file was divided into 89 partitions.

Dask uses lazy evaluation, which means that operations are organized into a task graph before the computation starts. The calculation was executed using the `compute()` method.

The execution time was 101.83 seconds, and the RAM usage at the end of execution was 0.30 GB. The calculated mean was approximately 11.3450456.

6 Performance Comparison

Method	Time (s)	RAM (GB)
Standard Pandas	392.28	9.36
Pandas Chunking	138.42	0.12
Dask DataFrame	101.83	0.30

Dask achieved the shortest execution time, followed by Pandas Chunking. Standard Pandas was the slowest method. All three methods produced almost identical mean values.

The RAM measurements were taken at different points during execution, so they are not directly comparable as peak memory measurements.

7 Conclusion

This practical work showed that large CSV files can be processed using different approaches. Standard Pandas successfully loaded the dataset but took the longest time. Pandas Chunking reduced the observed memory usage, while Dask achieved the fastest execution time in this experiment.

Chunking and Dask are useful techniques for working with large datasets, especially when memory resources are limited.